EXNO 2: Data Split up and Cross validation

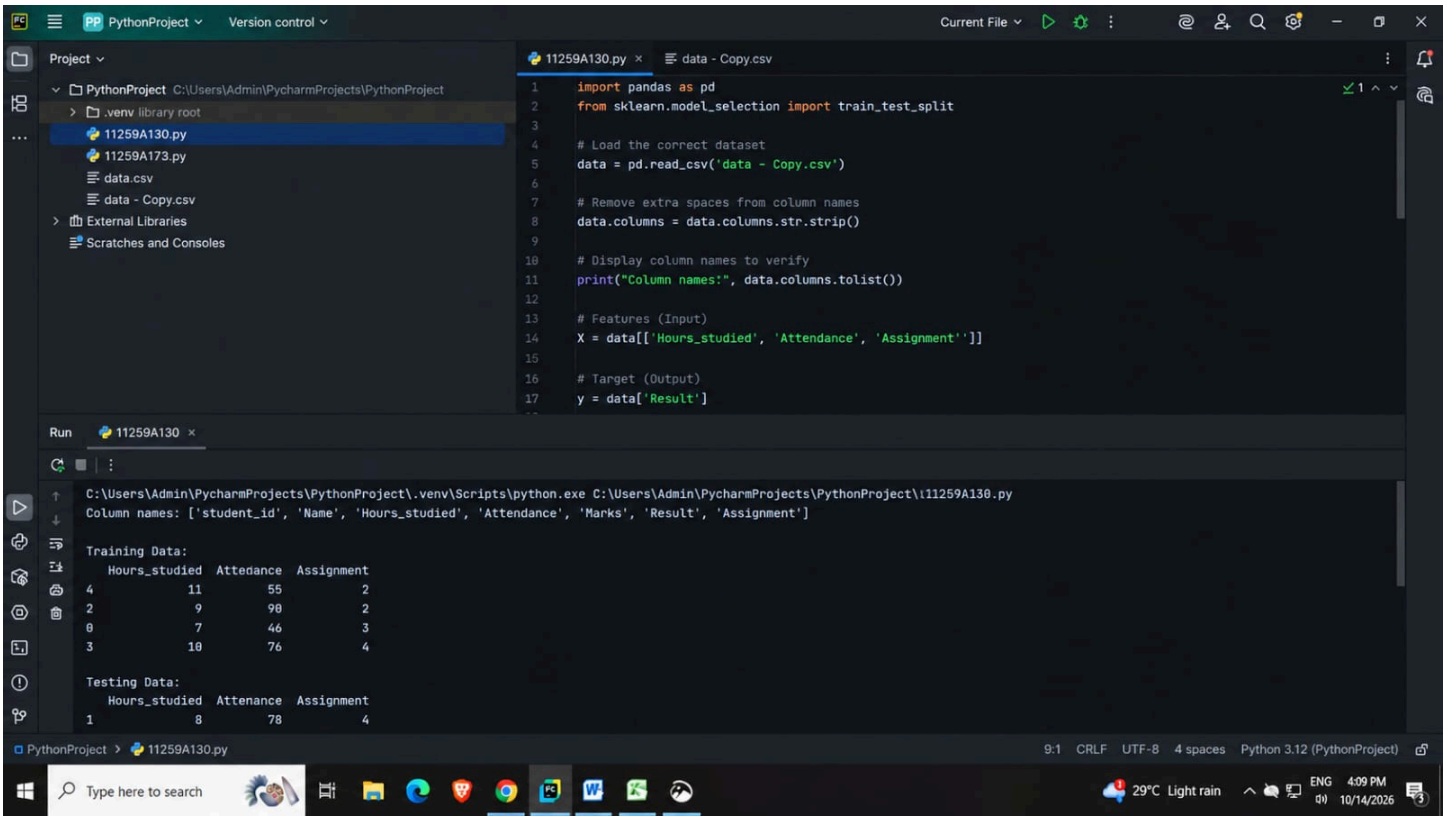
2.1 DataSplitting

Aim

To split a dataset into training and testing sets using the train_test_split() function from Scikit-learn, with 80% of the data used for training and 20% for testing.

Algorithm

1. Import the Pandas library and train_test_split from sklearn.model_selection.
2. Load the dataset from the CSV file data.csv using pd.read_csv().
3. Select the input features (Hours_Studied, Attendance, and Assignments) and store them in X.
4. Select the target output column (Result) and store it in y.
5. Split the dataset into training and testing sets using train_test_split() with test_size=0.2 and random_state=42.
6. Store the resulting data in X_train, X_test, y_train, and y_test.
7. Display the training data.
8. Display the testing data.
9. Print the number of training and testing samples.



Result

The dataset was successfully divided into training and testing sets using the train_test_split() function.

- **Training data:** 80% of the dataset.
- **Testing data:** 20% of the dataset.
- **Input features:** Hours_Studied, Attendance, and Assignments.
- **Target variable:** Result.

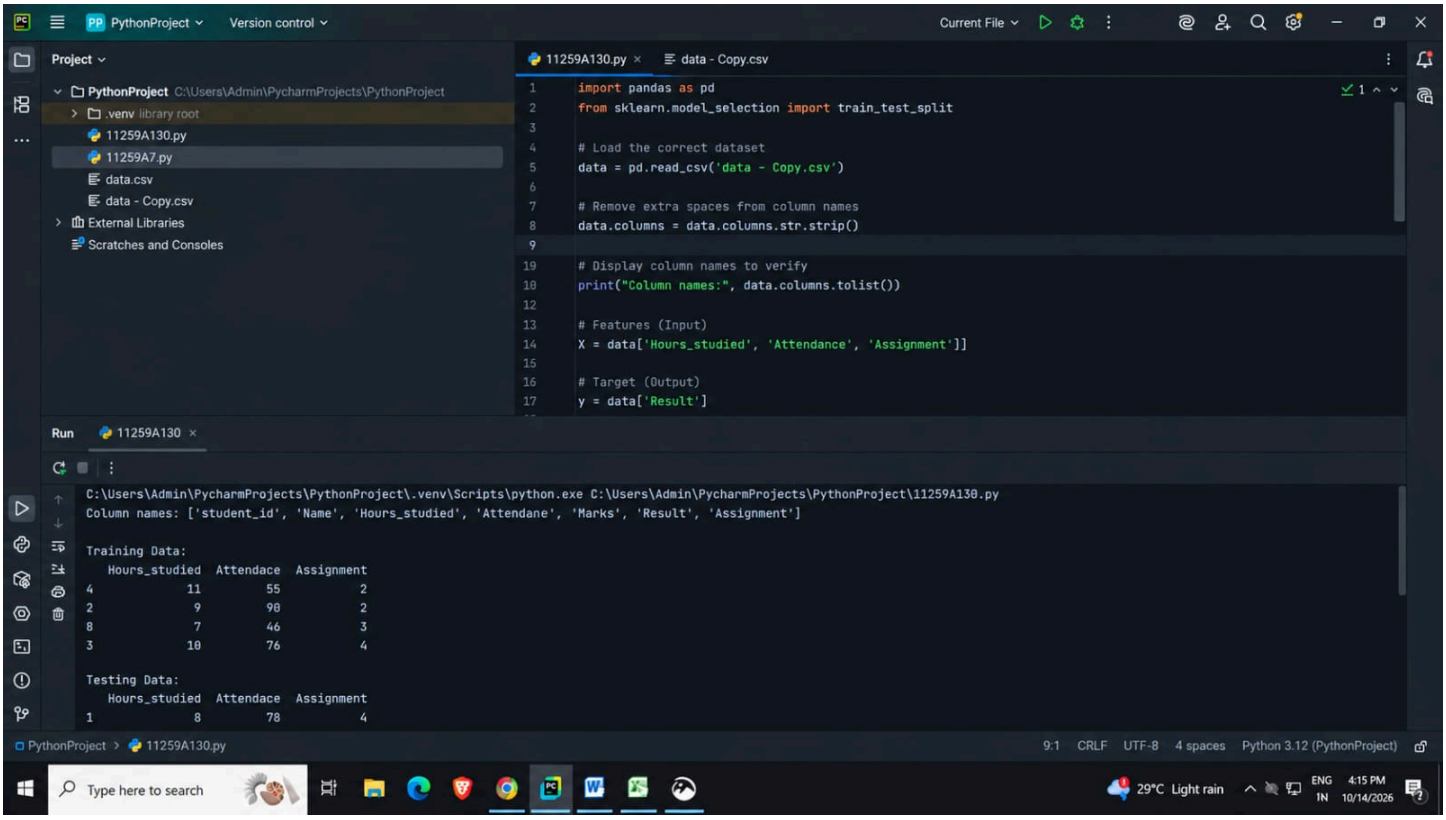
Data Split up and Cross validation.

Aim

To evaluate the performance of a K-Nearest Neighbors (KNN) classification model using 5-fold cross-validation and calculate its average accuracy using Scikit-learn.

Algorithm

1. Import the Pandas library and the required functions and classes from Scikit-learn.
2. Load the dataset from the CSV file data.csv.
3. Select the input features (Hours_Studied, Attendance, and Assignments) and store them in X.
4. Select the target variable (Result) and store it in y.
5. Split the dataset into training and testing sets using train_test_split() with 80% training data and 20% testing data.
6. Display the sizes of the training and testing datasets.
7. Create a KNN classification model using KNeighborsClassifier(n_neighbors=3).
8. Perform 5-fold cross-validation using cross_val_score(model, X, y, cv=5).
9. Display the accuracy score obtained for each of the five folds.
10. Calculate and display the average accuracy using scores.mean().



Result

The program successfully loaded the dataset, divided it into training and testing sets, and evaluated the KNN classifier using 5-fold cross-validation.

The output displays:

- The number of training samples.
- The number of testing samples.
- The accuracy scores for each of the five folds.
- The average accuracy of the KNN model